

Meanings and Measurements: Multi-Agent Probabilistic Grounding for Vision-Language Navigation

Abstract—Robots must convert instructions such as “go two meters to the right of the fridge” into goals that are both semantically correct and metrically executable. We present MAPG, a multi-agent probabilistic grounding framework that decomposes each instruction into anchor, directional, and metric clauses, grounds them against an online 3D scene graph, and composes analytic kernels into a continuous planner-ready goal distribution. We introduce MAPG-Bench with 100 annotated queries across 41 HM3D scenes. In matched full-run comparisons, MAPG reduces object-to-world error from 4.30 to 2.92 m with GPT-5.6 Luna and from 4.72 to 2.57 m with Gemini 3.7 Flash. It also reduces object-to-object error from 7.03 to 2.09 m over the single-image spatial baseline. Ablations show that explicit spatial reasoning is more reliable than replacing it with chain-of-thought prompting, particularly under occlusion.

I. INTRODUCTION

Robots collaborating with humans must turn natural-language goals into grounded decisions that a planner can execute. These goals often mix semantics and measurement: “two meters to the right of the fridge” requires resolving the correct fridge, interpreting “right” in a consistent frame, and placing a waypoint at the requested distance. We call these *metric-semantic* instructions. Their intended output is not only an object label or discrete action, but a continuous region in the world anchored to a grounded object pose.

Modern vision-language navigation and embodied question-answering systems reason over egocentric observations and persistent representations such as 3D scene graphs [1], [2]. However, many still collapse grounding into a single target or action. Single-image spatial specialists can estimate relations in one view but cannot accumulate evidence across viewpoints [3]. Scene-graph systems preserve world state, yet do not directly enforce metric offsets or compose several spatial constraints. These limitations matter under partial observability, repeated object instances, and ambiguous frames of reference.

We propose **MAPG**, a Multi-Agent Probabilistic Grounding framework. MAPG separates referent grounding, directional reasoning, and metric estimation, then combines their outputs as analytic distributions in physical space. This provides a common interface between specialist VLMs, an online scene graph, and a motion planner. Our contributions are: (1) a compositional 3D grounding framework that produces planner-ready goal distributions; (2) MAPG-Bench, with 100 annotated metric-semantic queries across 41 HM3D scenes [4]; and (3) matched comparisons and

ablations showing lower geometric error, stronger anchor selection, and improved robustness under occlusion.

II. RELATED WORK

Embodied QA and vision-language navigation connect language to actions under partial observability [1], [5], while 3D scene graphs provide persistent metric and semantic memory for long-horizon reasoning [2], [6]. Spatial specialists such as SpatialRGPT reason about geometric relations from images [3], but a single view does not provide a persistent allocentric goal. LINGO-Space represents referred space as a probability distribution and updates it from composite expressions [7]. MAPG builds on these directions by grounding against an online 3D scene graph, separating anchor, direction, and metric estimation, and composing the resulting distributions into a waypoint that can be queried directly by a planner.

III. METHOD

Problem. Given an instruction q , egocentric observations, and a partial 3D scene graph $\Gamma = (V, E)$, MAPG handles two outputs. Object-to-world (O-W) queries require a density $P(x | q, \Gamma)$ over allocentric waypoints $x \in \Omega_{\text{free}}$. Object-to-object (O-O) queries require selecting the object instance that satisfies a relation. In both cases, the output must agree with the referent, spatial predicate, and metric constraint.

Decomposition and grounding. The Orchestrator parses q into spatial description clauses [8]: an anchor, relation, optional metric distance, and composition rule. For the running instruction, it returns `fridge`, `intrinsic:right`, and `2.0m`. Multi-clause instructions produce one clause per constraint. The Orchestrator also determines whether a direction is camera-relative or intrinsic to the anchor, which fixes the coordinate frame before metric reasoning begins.

The Grounding Agent resolves each anchor against Γ . It matches label equivalents such as “couch” and “sofa,” ranks candidate instances using visual evidence from multiple views and a salience prior, and maintains a belief $B_t(o)$ over candidates. The Verifier rejects candidates unsupported by the image or scene graph and records structured feedback. Rejected instances are excluded from the next pass, while new observations can raise the belief of an object that was initially occluded. The agent therefore commits only after comparing evidence across the trajectory instead of selecting the first matching scene-graph node.

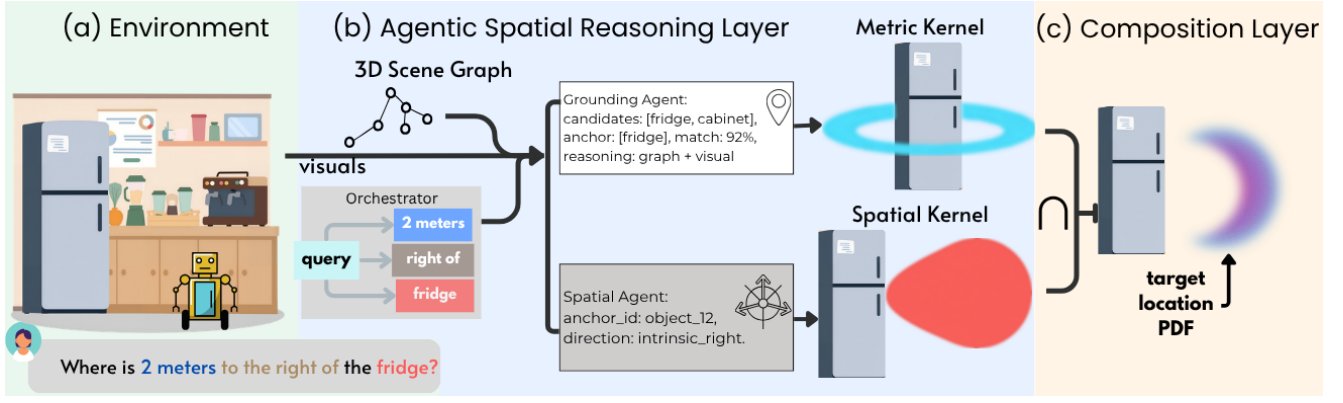


Fig. 1. **MAPG overview.** The robot receives a spatial query and builds a 3D scene graph. The Orchestrator extracts anchor, relation, and metric clauses; specialist agents ground the anchor and predict analytic kernel parameters; kernel composition produces a continuous target-location distribution for planning.

Analytic kernels. For a resolved anchor pose t_o , MAPG instantiates semantic, metric, and directional log-kernels:

$$\ell_{\text{sem}}(x) = -\frac{\|x - t_o\|^2}{2\sigma_s^2}, \quad \ell_{\text{met}}(x) = -\frac{(\|x - t_o\| - d_0)^2}{2\sigma_m^2}, \quad (1)$$

$$\ell_{\text{dir}}(x) = \kappa(R_o m(\theta_0, \phi_0))^\top (\widehat{x - t_o}). \quad (2)$$

The VLM predicts a camera-relative direction from the query, image, pose, and scene-graph context. Given agent yaw ψ , MAPG converts this direction to the world frame before evaluating the kernel. Metric distance d_0 comes from the parsed instruction, while the coordinate transform and tolerance are enforced analytically. For predicates marked as intrinsic, R_o uses the stored anchor orientation rather than the current camera frame. The VLM estimates the uncertain visual direction, while geometry handles the exact offset and frame conversion.

Composition, verification, and planning. Clause-level kernels combine as a Product of Experts [9], $\log P(x) = \sum_j (\ell_{\text{sem}}^j + \ell_{\text{met}}^j + \ell_{\text{dir}}^j)$, normalized over free navigable space. Compound instructions instantiate one kernel triplet per clause, so the final peak must satisfy all active constraints. Before planning, the Verifier checks the parsed logic, visual support for the anchor, directional agreement with the observation, and semantic label compatibility. A failed check returns control to the responsible agent. The navmesh-masked highest-density point becomes the planner goal only after verification. This keeps uncertainty through grounding while returning a single allocentric waypoint when planning requires one.

IV. MAPG-BENCH

MAPG-Bench isolates metric-semantic grounding that is not measured by single-image spatial QA or semantic embodied QA. It contains 100 annotated queries across 41 HM3D indoor scenes rendered in Habitat-Sim. Object-to-world queries require a continuous allocentric waypoint relative to an anchor. Object-to-object queries require selecting an object from three accepted instances around the target. The two query types separate waypoint accuracy from

relational object selection while using the same mapped environments.

Annotators explored each scene, selected the intended anchor instance, and placed the target on a navigable surface. Each record stores the scene and start state, query, anchor instance, egocentric observations, and either the target coordinate or accepted object set. The query set covers left, right, front, behind, above, below, near, between, from, and towards relations, with metric offsets from 0.5 to 4.0 m. An intrinsic-frame subset tests whether front, behind, left, and right are interpreted from the anchor’s orientation rather than the camera view. We use fixed scene starts and the same queries for all methods. This keeps viewpoint and exploration difficulty fixed when the backend or grounding pipeline changes.

V. EXPERIMENTAL RESULTS

Setup and metrics. We compare MAPG with SpatialRGPT [3] and an adapted GraphEQA [2] planner interface. SpatialRGPT receives the strongest selected egocentric view. GraphEQA receives the persistent scene graph and emits a waypoint through the same planner-facing record format as MAPG. MAPG variants share the same pipeline and differ only in VLM backend. All methods are evaluated by one adapter over saved outputs.

O-O error is the conditional mean minimum 3D distance between predicted and accepted object instances. O-W error is the conditional mean horizontal waypoint error. Combined angular error measures the difference between predicted and target anchor-to-goal rays. Object selection is correct when the predicted three-object set intersects the accepted set. We also report intrinsic-relation accuracy, valid-waypoint completion, exact anchor-instance accuracy, and stored trajectory length. Completion only checks that the system returned a navigable waypoint; it is not a correctness metric.

Metric-semantic grounding. With GPT-5.6 Luna fixed as the shared backend, MAPG lowers O-W error from 4.30 to 2.92 m, a 32.1% reduction, and angular error from 87.46° to 69.16°. Exact anchor accuracy rises from 0.20 to 0.61, and completion rises from 0.27 to 0.93. The matched Gemini 3.7

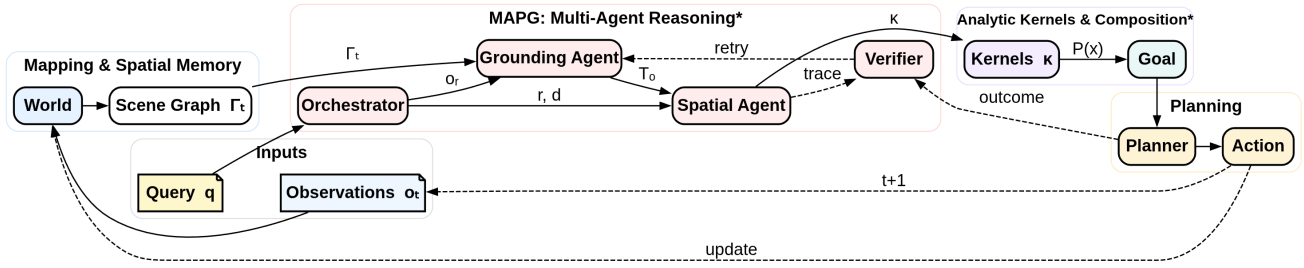


Fig. 2. **MAPG system flow.** The online scene graph and observations provide persistent spatial memory. The Orchestrator assigns structured subproblems to the Grounding Agent and Spatial Agent, while the Verifier can request another grounding pass. Analytic kernels compose into a goal distribution whose navmesh-masked maximum is sent to the planner.

Method	O-O ↓ (m)	O-W ↓ (m)	Angle ↓ (deg.)	Obj. Sel. ↑	Intrinsic ↑	Completion ↑	Anchor ↑	Traj. ↓ (m)
SRGPT (VILA1.5-8B)	7.03	6.87	81.05	0.00	0.61	0.95	N/A	N/A
GraphEQA (GPT-5.6 Luna)	4.16	4.30	87.46	0.04	0.08	0.27	0.20	14.96
GraphEQA (Gemini 3.7 Flash)	4.51	4.72	80.77	0.05	0.52	0.81	0.40	12.33
MAPG (GPT-5.2)	2.45	2.92	72.93	0.22	0.59	0.86	0.56	4.24
MAPG (Gemini 3.7 Flash)	2.13	2.57	73.87	0.27	0.53	0.83	0.57	5.38
MAPG (Claude Opus 4.6)	2.09	2.66	73.18	0.26	0.47	0.81	0.55	4.33
MAPG (Claude Sonnet 5)	2.17	2.59	76.59	0.24	0.47	0.74	0.51	6.64
MAPG (GPT-5.6 Luna)	2.53	2.92	69.16	0.23	0.67	0.93	0.61	4.29

TABLE I

MAPG-BENCH RESULTS. O-O, O-W, AND TRAJECTORY LENGTH ARE IN METERS; ANGLE IS IN DEGREES. COMPLETION RECORDS WHETHER A VALID FINAL WAYPOINT IS RETURNED AND IS NOT A CORRECTNESS METRIC. BOLD MARKS THE BEST VALUE IN EACH COLUMN.

Flash comparison shows the same pattern: O-W error falls from 4.72 to 2.57 m. The gain is not explained by backend strength alone because the scene-graph baseline and MAPG use the same VLM in both comparisons.

For O-O queries, MAPG with Claude Opus 4.6 reaches 2.09 m error compared with 7.03 m for SRGPT, a 70.3% reduction. A selected image can contain a plausible chair while missing another instance that better satisfies the relation. MAPG first resolves the anchor in the persistent map, then evaluates all accepted candidate objects relative to that 3D pose. This changes the problem from a view-dependent estimate to a map-consistent selection.

Backend and task transfer. The best values occur on different backends: Gemini Flash gives the lowest O-W error, Opus gives the lowest O-O error, and Luna gives the lowest angular error and highest anchor accuracy. The pipeline exposes these differences without changing its composition interface. On the intrinsic-frame subset, Luna improves from 0.08 with GraphEQA to 0.67 with MAPG. Flash changes only from 0.52 to 0.53, showing that decomposition helps most when the backend can estimate the anchor’s orientation from the available views. Table II evaluates transfer to HM-EQA, where explicit metric constraints are absent.

MAPG with Opus reaches 0.71 answer accuracy compared with 0.64 for matched GraphEQA and 0.44-0.54 for the reported Explore-EQA variants. The same configuration also has the shortest stored trajectory. MAPG-Bench remains the

Method	Accuracy ↑	Traj. (m) ↓
Explore-EQA, Llama4-Mav	0.44	10.4
Explore-EQA, Gemini 2.5 Pro	0.54	12.3
GraphEQA, GPT-5.2	0.63	7.1
GraphEQA, Claude Opus 4.6	0.64	7.4
MAPG, GPT-5.2	0.60	6.9
MAPG, Claude Opus 4.6	0.71	6.6

TABLE II

ANSWER ACCURACY AND TRAJECTORY LENGTH ON HM-EQA [5].

Method	Full ↑	Occluded ↑
GraphEQA	0.34	0.30
MAPG, CoT without spatial reasoner	0.20	0.30
MAPG, explicit spatial reasoner	0.42	0.50

TABLE III

OBJECT-SELECTION SUCCESS IN THE SPATIAL-REASONING ABLATION. EXPLICIT REASONING IS STRONGEST ON THE FULL SET AND UNDER OCCLUSION.

primary metric-semantic evaluation, while HM-EQA checks that decomposition and evidence accumulation do not trade away semantic QA performance.

Ablation. Replacing the spatial reasoner with a single chain-of-thought planner reduces object selection from 0.42 to 0.20. The prompted planner sees the same query and

observations but must produce the relation and target in one free-form step. Under occlusion, explicit reasoning reaches 0.50 while both GraphEQA and the CoT variant remain at 0.30. Maintaining an intermediate anchor belief and revisiting rejected candidates matters when the relevant object is not visible in the first view. The ablation therefore isolates structured spatial inference, rather than additional observations or a stronger backend.

VI. DISCUSSION

What improves. The results separate two effects. First, multi-view evidence improves instance grounding. Repeated objects are common in HM3D, and the closest visible instance is not always the one named by the query. The Grounding Agent can reject that instance and continue from another view, which explains the higher anchor accuracy and shorter trajectories in the matched comparisons. Second, analytic composition improves geometric consistency because direction and distance are enforced together rather than left inside a free-form waypoint prompt. For the running fridge example, GraphEQA can return a plausible region to the right of a fridge without enforcing the requested radius. MAPG first fixes the fridge instance, then intersects directional and radial constraints before selecting a navigable point. The resulting coordinate is expressed in the persistent world frame, so the planner does not have to reinterpret a textual answer.

What changes across backends. No single backend wins every column. Flash produces the lowest O-W error, Opus the lowest O-O error, and Luna the best angular and anchor results. These differences are useful rather than contradictory. Referent resolution, intrinsic-frame reasoning, and structured metric parsing place different demands on the VLM. MAPG keeps those effects localized to the responsible agent while the analytic composition and evaluation protocol remain fixed.

Interpreting the interface and metrics. Completion records whether a navigable output exists, even when the selected anchor or waypoint is wrong. Conditional O-O and O-W errors measure geometry for resolved outputs, while anchor and object-selection accuracy expose association failures that distance alone can hide. We therefore read the columns jointly rather than treating completion as task success. MAPG retains a spatial density until planning instead of asking the VLM for a free-form coordinate. Each clause removes regions that violate one part of the instruction, and the navmesh removes occupied points from the remaining support. This makes an error traceable to anchor belief, frame selection, metric parsing, or obstacle masking. The current implementation returns the highest-density navigable point, while leaving the grounding interface independent of the downstream planner.

Failure modes. Errors remain when the scene graph never contains the relevant object, when an intrinsic direction cannot be inferred from partial views, or when a target lies close to a wall and the navmesh shifts the density maximum. In the first case, no grounding policy can select

an instance that never enters the map. In the second, “in front of the sofa” may be interpreted from the camera rather than the sofa’s facing direction when its orientation is visually unclear. In the third, the kernel peak can lie in occupied space, so masking selects the nearest navigable point and introduces geometric error. These failures point to better object-orientation estimates, uncertainty-aware mapping, and explicit treatment of obstacle-induced displacement rather than simply larger prompts or a single stronger backend.

Reproducibility. Every method is evaluated from its saved trajectory and prediction records with the same adapter, scene starts, and query set. The adapter retains missing outputs rather than silently dropping failed queries and computes all geometric measures in the registered scene frame. The project page provides benchmark examples, metric definitions, and additional results.

VII. CONCLUSION

MAPG turns metric-semantic language into continuous 3D goal distributions by separating anchor, directional, and metric reasoning and composing their outputs in physical space. Across MAPG-Bench, it improves geometric grounding over scene-graph and single-image baselines, while matched-backend comparisons separate pipeline gains from VLM choice. The ablations show that explicit spatial reasoning is important under ambiguity and occlusion, and the HM-EQA results show that the same interface remains competitive when metric constraints are absent. MAPG provides a practical interface between language understanding, persistent spatial memory, and motion planning for open-world navigation.

REFERENCES

- [1] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra, “Embodied question answering,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [2] S. Saxena, B. Buchanan, C. Paxton, B. Chen, N. Vaskevicius, L. Palmieri, J. Francis, and O. Kroemer, “Grapheqa: Using 3d semantic scene graphs for real-time embodied question answering,” *arXiv preprint arXiv:2412.14480*, 2024.
- [3] A.-C. Cheng, H. Yin, Y. Fu, Q. Guo, R. Yang, J. Kautz, X. Wang, and S. Liu, “Spatialrgpt: Grounded spatial reasoning in vision language models,” *arXiv preprint arXiv:2406.01584*, 2024.
- [4] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang, M. Savva, Y. Zhao, and D. Batra, “Habitat-matterport 3d dataset (hm3d): 1000 large-scale 3d environments for embodied ai,” 2021.
- [5] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh, “Explore until confident: Efficient exploration for embodied question answering,” 2024. [Online]. Available: <https://arxiv.org/abs/2403.15941>
- [6] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suen-derhau, “Sayplan: Grounding large language models using 3d scene graphs for scalable robot task planning,” 2023.
- [7] D. Kim, N. Oh, D. Hwang, and D. Park, “Lingo-space: Language-conditioned incremental grounding for space,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 9, 2024, pp. 10 314–10 322.
- [8] T. Kollar, S. Tellex, D. Roy, and N. Roy, “Toward understanding natural language directions,” in *Proceedings of the 5th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2010, pp. 259–266.
- [9] G. E. Hinton, “Training products of experts by minimizing contrastive divergence,” *Neural Computation*, vol. 14, no. 8, pp. 1771–1800, 2002.